Advanced Topics in Deep Learning

IE 663
Lecture 0

January 5, 2026.

1. Credit Requirements
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Credit Requirements

- **Course Project:** 35%
    - Topics will be floated soon - selection to be done **only** from the list of topics floated.
    - Team size: Limited to maximum of 2 members (strict limit). Single member teams encouraged.

Credit Requirements

Course Project: 35%

- Team details and interests will be collected immediately after the course drop deadline.
- Course project topic statements will be floated based on the interests provided and bids, topic preferences will be collected from teams on the floated topics.
- Based on the bids and topic preferences, course project topics will be allotted.
- Course project will have three evaluation phases:
  - Preparatory Assessment - 10 % of total project score
  - Mid-term Assessment - 35 % of total project score
  - Continuous Assessment - 10% of total project score
  - End-term Assessment - 45 % of total project score
- More details to be provided soon.
- Please talk to classmates and form teams (if inclined) based on mutual interests.

Credit Requirements

Course Project:

- Preparatory assessment - **10%** of total project score. Main components involve:
  - Teams understanding the tasks/goals outlined in the Course Project Topic Statement.
  - Teams exploring ideas for the project statement, finding datasets, data collection procedures, codebases, relevant research papers to read and carving out the main milestone timelines for the project.
  - Discussing and finalizing these by meeting with the TAs/Instructor.
  - Teams preparing a presentation containing details about the team's exploration and details of the initial reading (e.g. blogs/papers/other resources) toward the project and outlining the roadmap for the project with timelines, milestones, etc.
  - In-person meet with the Instructor and TAs to discuss the preparatory activities.
- **Note:** Clear instructions on these activities will be provided later.

Credit Requirements

Course Project:

- Mid-term assessment - **35%** of total project score. Main components involve:
  - Based on regular meetings of teams with TAs/Instructor teams are expected to successfully progress on the execution of their projects.
  - Teams will present their work done till middle of semester towards the successful execution of the course project with necessary demonstrations.
  - Teams will also submit an extensive report along with code files and code walkthrough videos.
- **Note:** Clear instructions on these activities will be provided later.

Credit Requirements

Course Project:

- Continuous assessment - **10%** of total project score. Main components involve the following:
  - Teams will regularly meet TAs/Instructor and update their progress of the project during office hours and after the lectures (if required).
  - Teams will discuss on the datasets, models, experimental settings, training/validation/testing methodologies they would be choosing for their projects and get them verified by TAs and Instructor.
  - Teams will clear doubts related to their projects.
  - All team members should be present for the meetings.
- **Note:** Clear instructions on these activities will be provided later.

Credit Requirements

Course Project:

- End-term assessment - **45%** of total project score. Main components involve the following:
  - Based on regular meetings of teams with TAs/Instructor teams are expected to successfully execute their projects to completion.
  - Teams will present their work done towards the successful execution of the course project with necessary demonstrations.
  - Teams will also showcase their work on novel ideas/tasks beyond the task outlined in the course project topic statement.
  - Teams will also submit an extensive report along with code files and code walkthrough videos.
- **Note:** Clear instructions on these activities will be provided later.

Credit Requirements

IMPORTANT:

- Each student must mandatorily participate in executing the course project.
- Each student must appear for preparatory assessment meetings, continuous assessment meetings, mid-term and end-term assessment meetings.
- Students who do not execute their course projects and those who do not appear for regular meetings will be awarded **Fail** grade. (No exceptions!)
- NOTE: No verification emails about your course projects during placements in Spring 2026 semester !

Credit Requirements

- **Mid-Term Exam:** 20%
  - Mid-term exam will be open notes and open scribes.
- **Quizzes:** 20%
  - 2 quizzes would be held.
  - Equal weights to both quizzes
  - Announcements of quiz date and timings will be made in class. Moodle postings on quiz timing and instructions will be made a day before the quiz.
  - All quizzes will be open notes and open scribes.
- **Homework problems, practice questions (both theory and coding related) will be provided regularly.** (Will not be graded!) (Discussions on homework and practice problems in Moodle are encouraged.)

Credit Requirements

- **Class participation and scribing:** 5%
  - Specific students will need to scribe contents discussed during the lecture.
  - Scribe template (in latex) will be posted soon.
  - Scribes will form the primary source of reading material for other students.
  - Hence scribes are expected to be thorough and comprehensive.
  - Scribes which point to additional materials will be very helpful.
  - Each scribe will be reviewed by the Instructor/TAs for completeness. Scribes which are not perfect will be sent back to the concerned participants for rework (which might span multiple iterations if required).

Credit Requirements

- **Research Understanding Presentations:** 10%
  - Each module to be covered during the course will be followed by a research understanding meeting.
  - Participants will bid for, read, understand and present recent research papers related to the contents discussed in class. Research Paper details will be posted during the module discussions.
  - Team composition for research understanding presentations will be decided later after drop deadline.
  - Non-presenting Participants will also be evaluated for taking part in the discussion during the presentations.
  - Excellent presentations will be noted, and will deserve a special mention. Grade advantages would be provided to those teams who present excellently.
- **Note:** Clear instructions on these activities will be provided later.

Credit Requirements

- **Challenge Assignment:** 10%
  - One challenge assignment will be provided after the mid-sem.
  - Students will be allowed to form 2-member teams. Individual participation is also allowed.
  - Students can propose solutions based on the problem description and solution requirements.
  - Submissions will be ranked based on specific considerations, which will be posted during the assignment announcement.
  - Students who provide best 3 solutions would be given extra marks leading to grade advantage, and their ideas will deserve special mention during the course.
- **Note:** Clear instructions on this challenge assignment will be provided later.

Essential Programming Skills

- Medium (or advanced) level knowledge of Pytorch is **essential** for the course.
- No special training for any programming tools will be provided.

Request to participants

- If you are completely new to Python
- If you do not have adequate exposure to Pytorch
- If you have not done any course on ML and/or DL
- If you are a B.Tech (or) B.S. sophomore
- If you have registered the course for audit
- If you have registered for another course in slot 12

Please de-register !!!

Request to participants

- Please bring a laptop and/or mobile phone to every class. We shall have several practice sessions through the course.
- Please keep your laptop speakers and mobile phones in mute mode.
- Please plan to be in class latest by 5:35 pm.
- Attendance will be monitored.
  - Regular participation in lectures, participation during research understanding presentations, project assessments (continuous, mid-term and end-term), scribing activities, etc. is essential to pass the course.
- However, no threshold on minimum expected attendance.

Course Outline

Tentative agenda:

- Theory behind Diffusion models, flow models and related aspects
- Neural operators - Design, theory and implementations
- Statistical learning theory ideas for Deep Neural Nets (e.g. PAC theory)
- Other recent topics
- Course Project will cover more topics which might not be discussed in class. Please make full use of course projects as they will be helpful for exploring other topics.

Alert about tagging

- Please do not perform arbitrary re-tagging of the course in asc.
- It leads to immediate cancellation of registration.
- Getting the course back in asc takes quite some effort (multiple visits to acad office, etc.)

Materials for self-study and Reference Texts

Materials for self-study

Lecture scribes prepared by participants and related research papers will be posted in Moodle.

Online tools

Please familiarize yourselves with the following web resources.

- Code management: `https://github.com`
- Model imports and exports: `https://huggingface.co`
- Model development, training and experimentation:
  - `https://www.kaggle.com/`
  - `https://colab.research.google.com/`

Teaching Assistants for the course

- Rahul Vaishnav (rahulvaishnav@iitb.ac.in)
- Bheeshm Sharma (bheeshmsharma@iitb.ac.in)

Office Hours

- Wednesdays, 12 PM to 1 PM.
- A regular venue will be confirmed by next week.

Note on academic integrity, copying and plagiarism

- As a responsible individual, each course participant will strictly adhere to the academic integrity principles laid out by the Institute.
- Any aberration will not be tolerated and will be dealt with appropriately.